

2D Electron Transport in Complex Geometries

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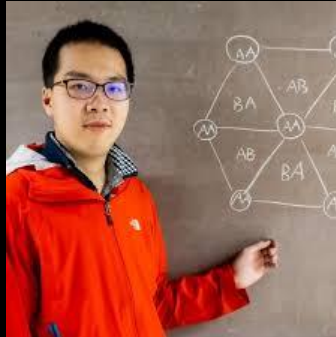


University of Colorado **Boulder**

Acknowledgments



Andy Lucas
(Boulder)



Canxun Zhang



Ludwig Holleis



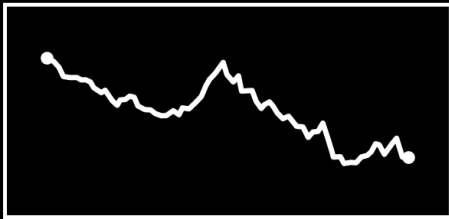
Rogue
(Cat)



Andrea Young
(UCSB)

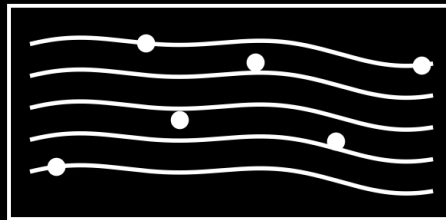
Overview

- Recall (from the previous talk) that current flow patterns (transport) are sensitive to the **type** and **strength** of different physical processes



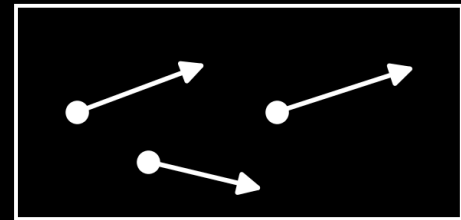
Diffusive

$$\gamma_{mr}$$



Hydrodynamic

$$\gamma_{mr} \ll \gamma_{mc}$$



Ballistic

$$\gamma_{mr} = \gamma_{mc} = 0$$

- But real materials often operate in *intermediate* regimes
- How does one use transport to determine parameters of this kind *quantitatively*?
- Imaging is awesome, but it takes a long time and feels like overkill to determine $O(1)$ phenomenological parameters...

Current fractions in a multiterminal geometry

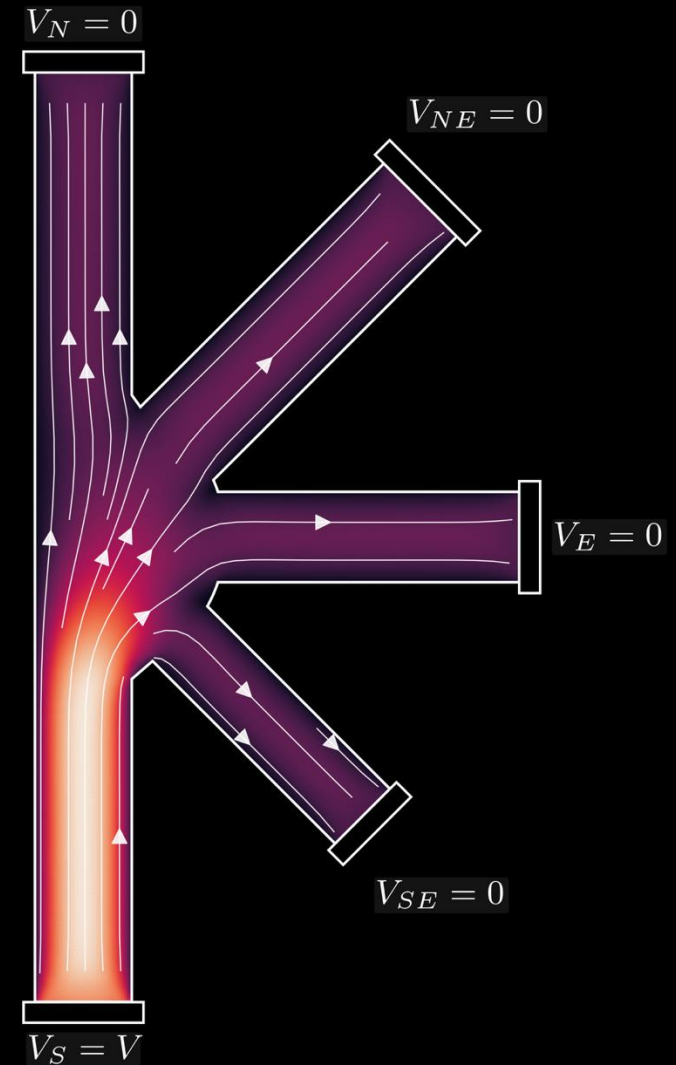
[JF, Lucas, [2605.03030](#)]

- Ballistic v.s. hydrodynamic, diffusive?
 - Ballistic carriers travel in semiclassical trajectories, so I_N should be large
- Hydrodynamic vs diffusive?
 - Recall expectations for resistance for rectangle with ℓ, w

$$R_{diff} \sim \gamma_{mr} \frac{\ell}{w}$$

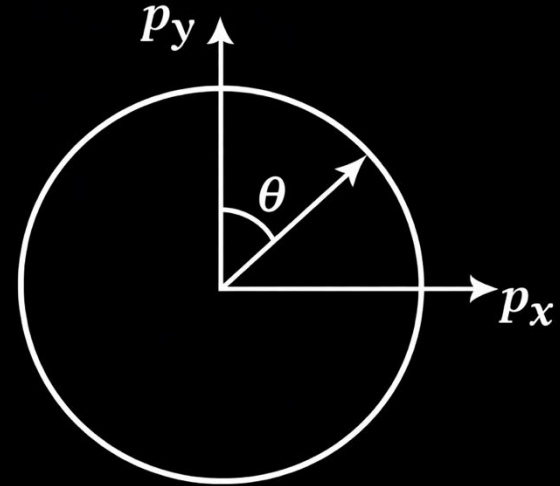
$$R_{hydro} \sim \gamma_{mc}^{-1} \frac{\ell}{w^3}$$

- So create channels with fixed $\frac{\ell}{w}$ but varying w !



A “toy” kinetic theory

- Linear Boltzmann equation
- Take fermions with $\varepsilon(|p|)$
(circular Fermi Surface)



- They have a distribution function $f(\mathbf{x}, \mathbf{p}, t)$, which in equilibrium and near $T = 0$ is $f_0 = \Theta(\epsilon_F - \varepsilon)$

$$\partial_t f + \mathbf{v}(\mathbf{p}) \cdot \nabla f = \mathcal{C}[f]$$

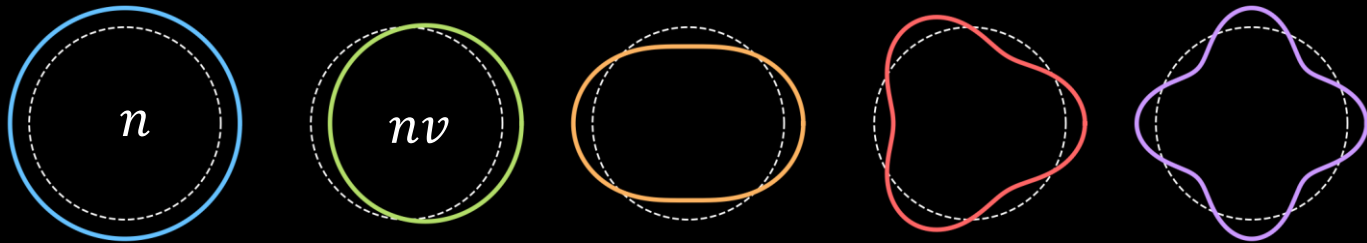
- Then parameterize deviation from equilibrium:

$$\delta f = \phi(\mathbf{x}, t, \theta) \delta(\epsilon_F - \varepsilon)$$

$$\mathcal{C} \rightarrow W[\phi(\theta)]$$

Harmonics capture conserved quantities:

$$\phi(\mathbf{x}, t, \theta) = \frac{a_0}{2} + \sum_n [a_n \cos n\theta + b_n \sin n\theta]$$



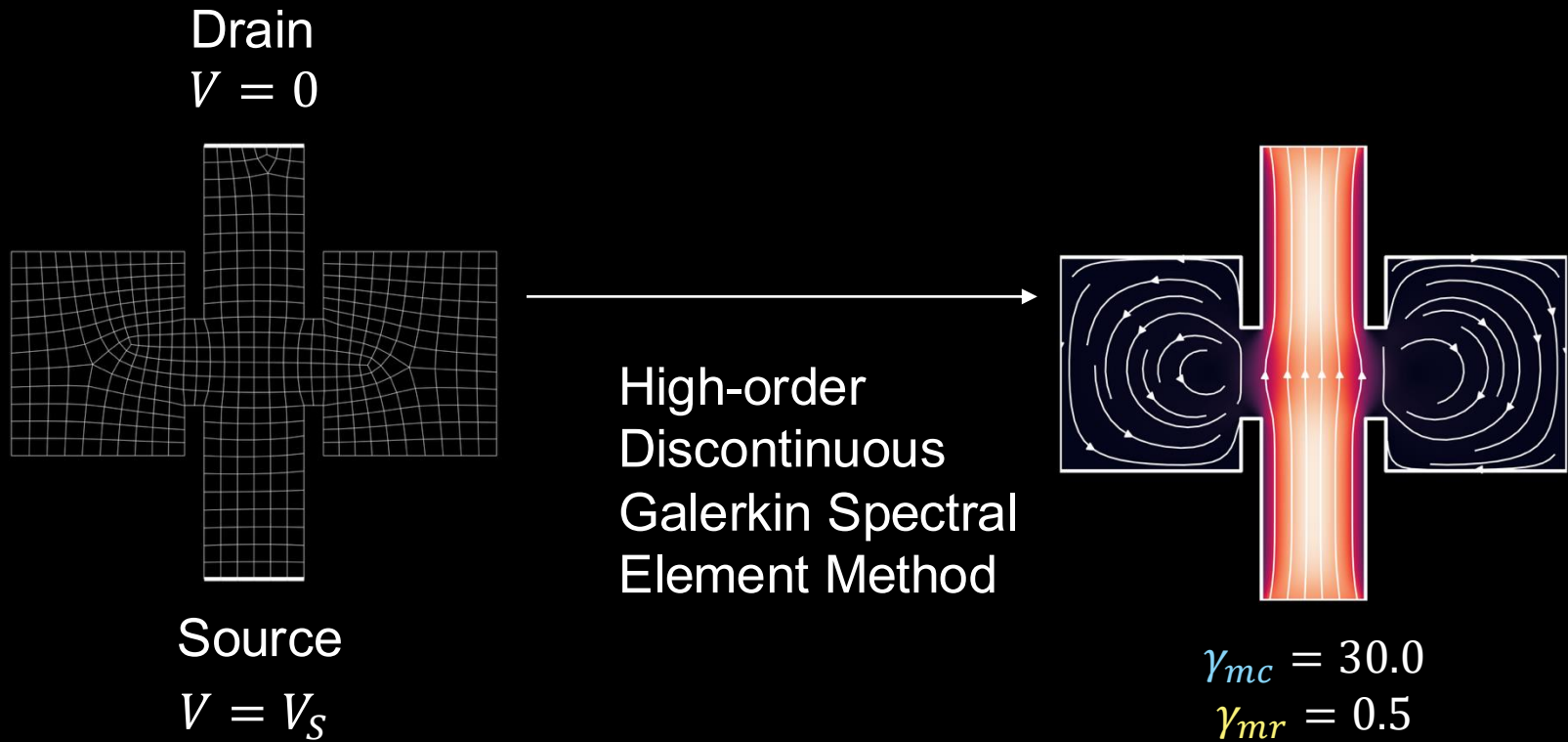
- Boltzmann becomes a tower of coupled PDEs

$$\partial_t a_0 + v_F (\partial_x a_1 + \partial_y b_1) = 0$$

$$\partial_t a_1 + \frac{v_F}{2} (\partial_x (a_2 + a_0) + \partial_y (b_2 - b_0)) = -\gamma_{\text{mr}} a_1$$

$$\partial_t a_n + \frac{v_F}{2} (\partial_x (a_{n+1} + a_{n-1}) + \partial_y (b_{n-1} - b_{n+1})) = -(\gamma_{\text{mr}} + \gamma_{\text{mc}}) a_n$$

Solve numerically in any geometry you want!



- Have studied wall boundary conditions **diffuse**, **specular**, or a **blend**
- Available at github.com/jackhfarrell/FermiSea.jl

[Zhang, Redekop, Stoyanov, JF, et al. [2603.11175](#)]

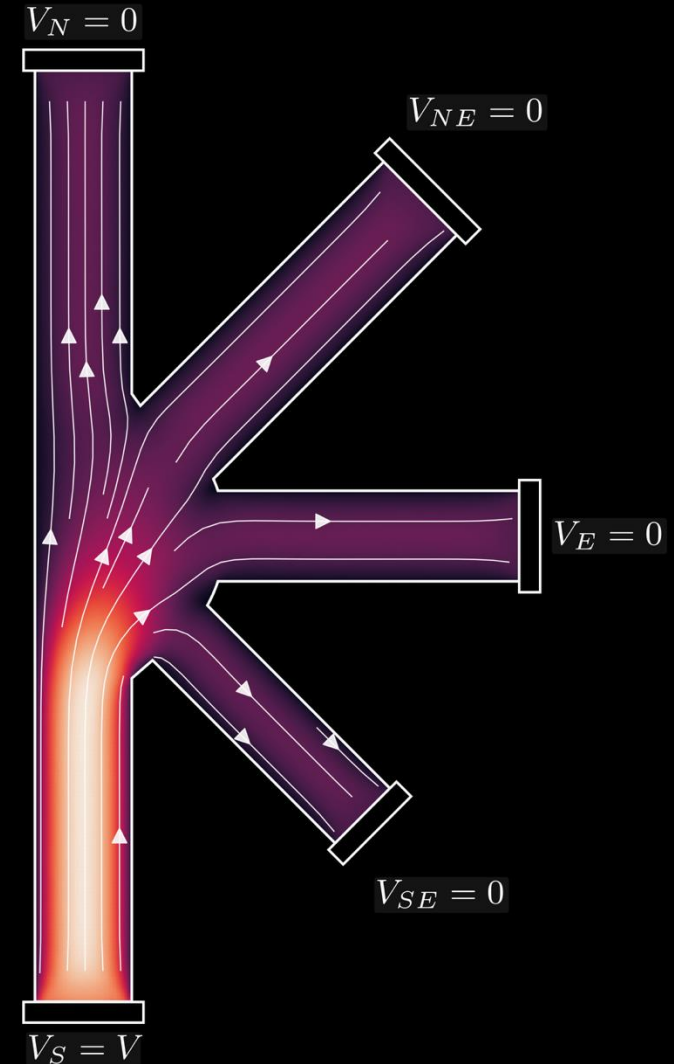
Reminder: Current fractions in a multiterminal geometry

- Ballistic v.s. hydrodynamic, diffusive?
 - Ballistic carriers travel in semiclassical trajectories, so I_N should be large
- Hydrodynamic vs diffusive?
 - Recall expectations for resistance for rectangle with ℓ, w

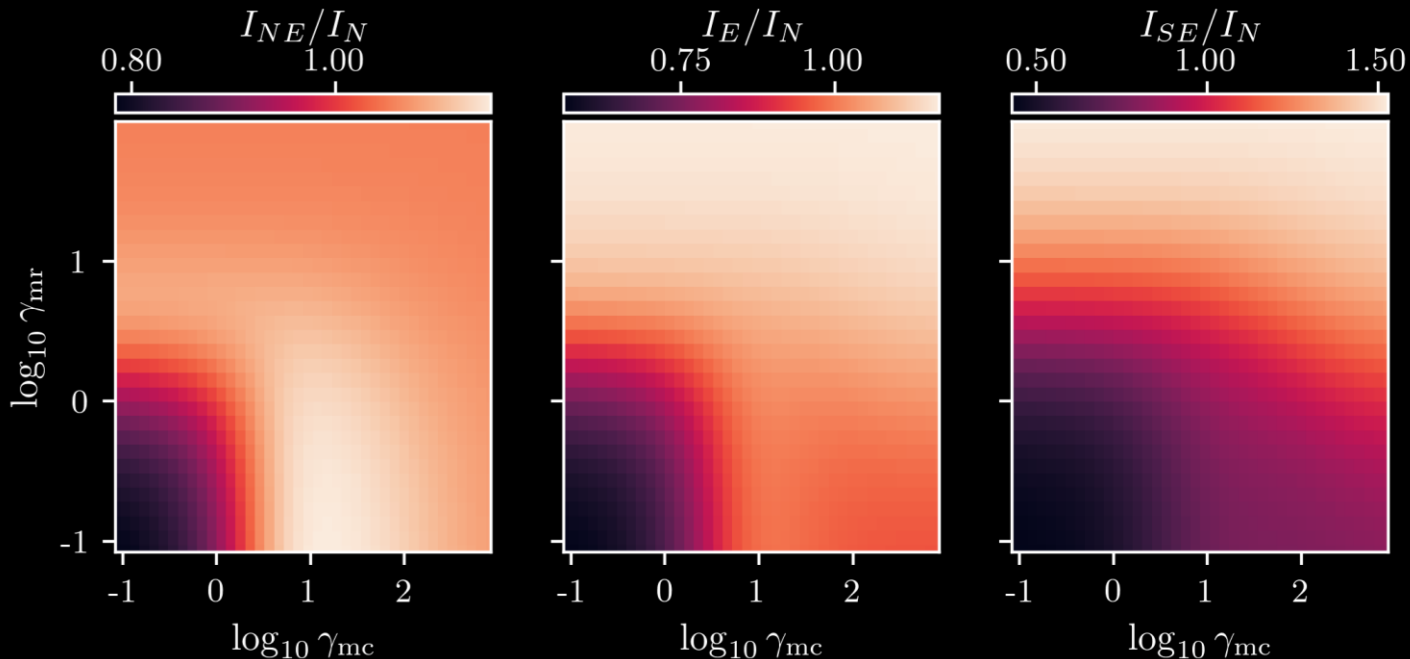
$$R_{diff} \sim \gamma_{mr} \frac{\ell}{w}$$

$$R_{hydro} \sim \gamma_{mc}^{-1} \frac{\ell}{w^3}$$

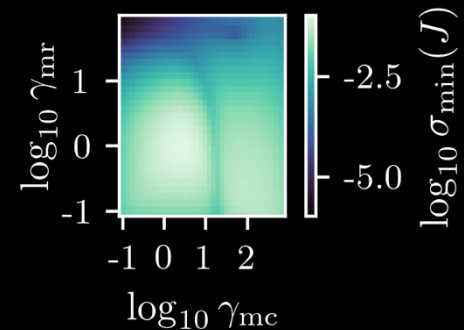
- So create channels with fixed $\frac{\ell}{w}$ but varying w !



Ballistic-Hydrodynamic-Diffusive Crossovers



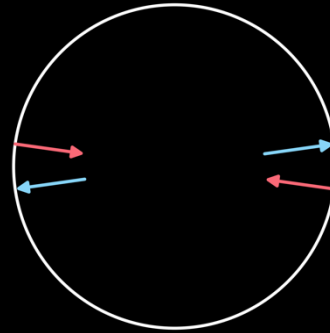
- If we were given a precise measurement of these three fractions, could we infer γ_{mr}, γ_{mc} ?
- Take take singular values of Jacobian of map $\gamma_{mr}, \gamma_{mc} \mapsto I_{NE}, I_E, I_{SE}$



Electron-electron collisions

- If our momentum-conserving collisions are short-range electron-electron scattering near $T = 0$, the collision integral $W[\phi]$ has very nice structure!

- Consider $\mathbf{p}_1, \mathbf{p}_2 \rightarrow \mathbf{p}_3, \mathbf{p}_4$, all four should live on FS, and conserve momentum and energy?

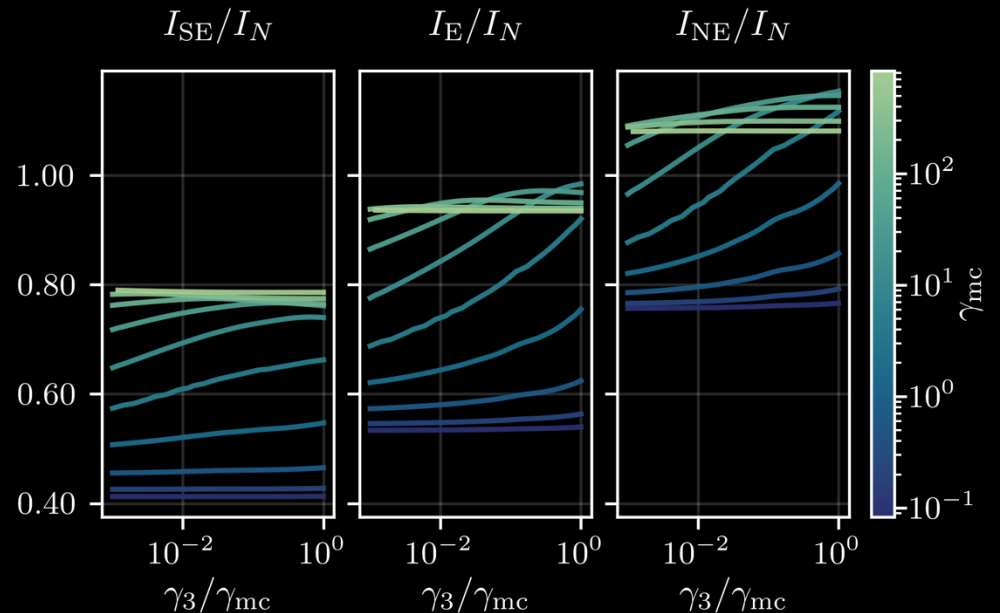
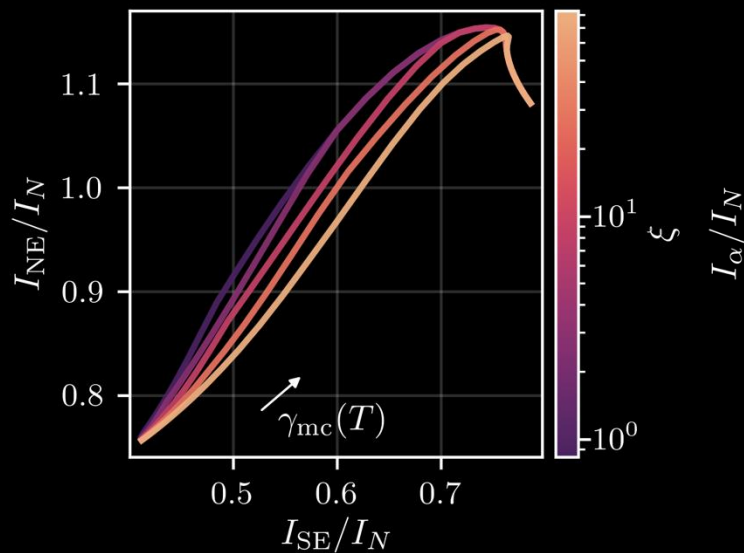


- Inversion symmetry: $W[\phi(\theta) - \phi(\pi - \theta)] \sim 0$
 $W[\phi(\theta) + \phi(\pi - \theta)] \sim \gamma_{mc}$
- Slightly better: remember this Fourier expansion?

$$\gamma_m = \begin{cases} \gamma_{mc} & m > 1 \text{ even} \\ \gamma_3 \left(\frac{m}{3}\right)^4 & m > 1 \text{ odd} \end{cases}$$

Sensitive to mode-resolved decay rates

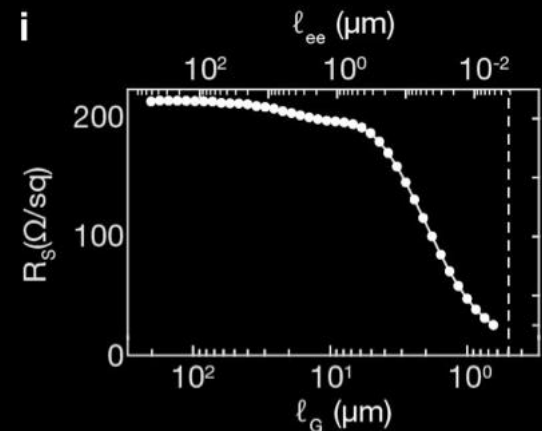
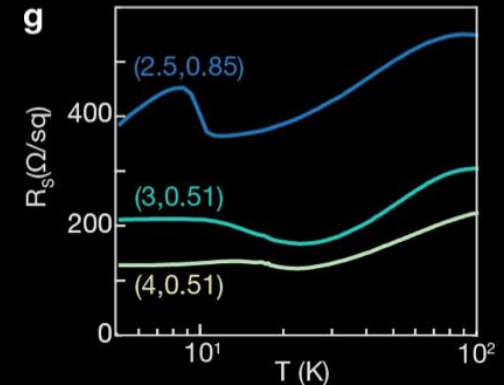
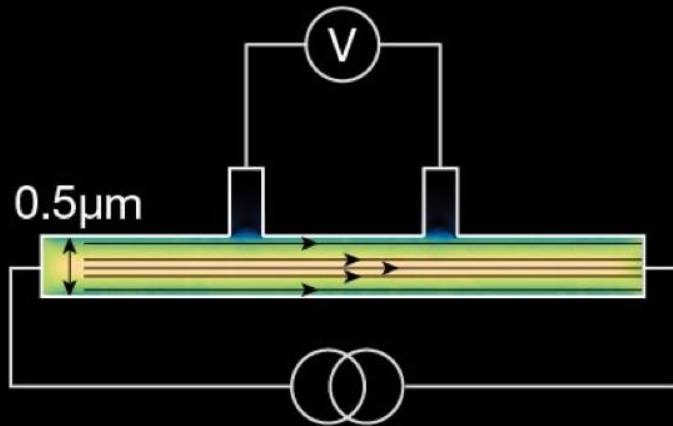
- Expectation: $\gamma_{mc} \sim T^2$, $\gamma_3 \sim T^4$
- Parameterize $\gamma_3 \sim \xi \gamma_{mc}$
- Imagine sweeping T



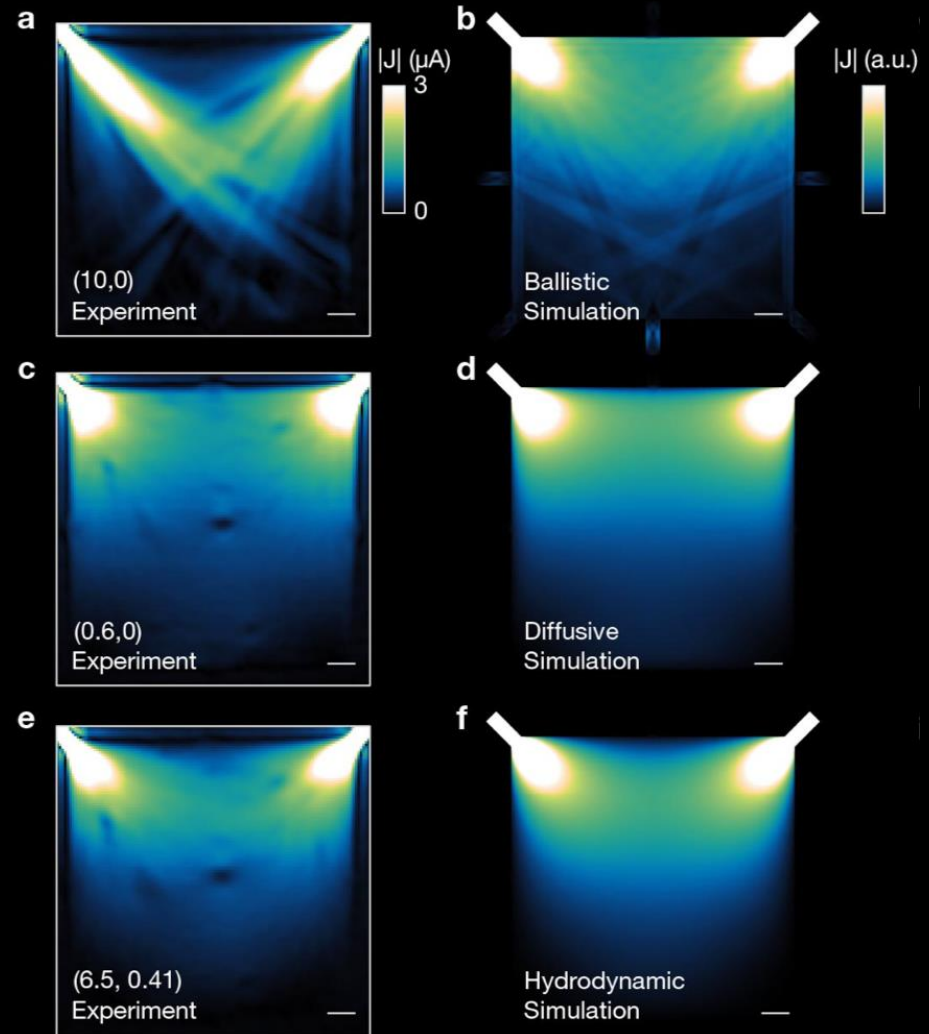
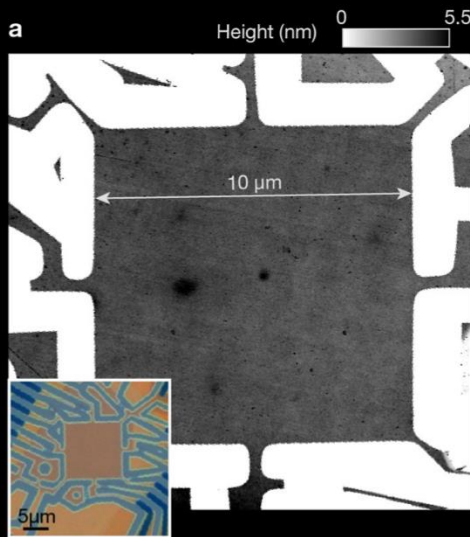
Proof of concept?

Holleis, Choi, Zhang, JF, *et al.* [2604.21912](#)]

- How are we doing in terms of modeling these transport observables?
- Extremely large device, 13 layer rhombohedral stacked graphene!



Proof of concept in even bigger device?



Future direction: anisotropic FS?

- Imagine single Fermi surface of some arbitrary shape parameterized by arclength s

$$\delta f = \phi(\mathbf{x}, t, \mathbf{p}) \delta(\epsilon_F(s) - \varepsilon)$$

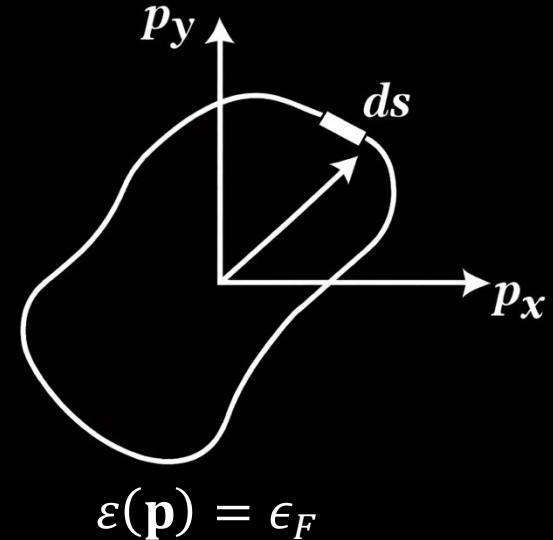
- Want to expand ϕ in functions

$$\phi(\mathbf{x}, t, s) = \sum_n a_n(\mathbf{x}, t) \chi_n(s)$$

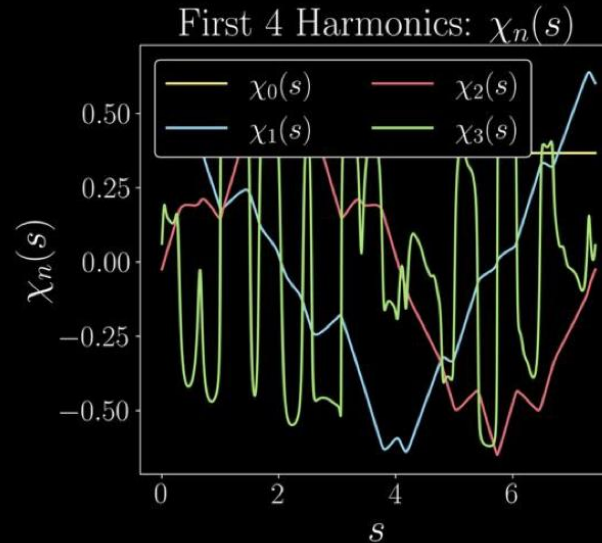
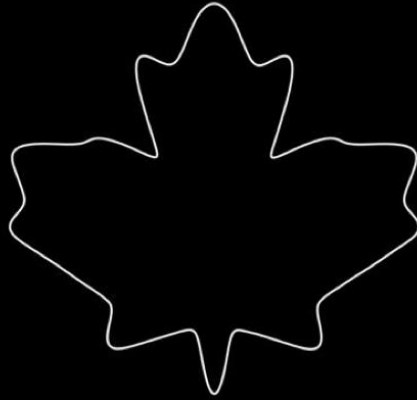
$$\begin{aligned} \int \frac{d^2 \mathbf{p}}{(2\pi)^2} \chi_n(s) \chi_m(s) \delta(\epsilon_F(s) - \varepsilon) &= \int \frac{ds}{(2\pi)^2} \chi_n(s) \chi_m(s) \frac{1}{v_F(s)} \\ &= \delta_{nm} \end{aligned}$$

- Boltzmann becomes a vector advection equation:

$$\partial_t \mathbf{a} + \partial_x (L_x \mathbf{a}) + \partial_y (L_y \mathbf{a}) = \mathbf{W}[\mathbf{a}]$$



Example / outlook



- Should be very interesting to try and model space resolved transport in materials w/ more complicated FS (including honest collision integral, see the methods of next talk!)
- In particular, can you engineer geometries in 2D (or 3D) to try and measure whichever parameter (rotational viscosity?)
 - inverse design?

Thanks for your attention! 😊